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1.0 GENERAL

1.1 Scope

1.1.1 This Specification outlines the general requirements for the protection against corrosion of the external surfaces of ferrous metal and low alloy (less than 12% Cr) steels using protective coating systems. This Specification also addresses the requirements for protection against corrosion of insulated and un-insulated austenitic stainless steel from corrosion.

1.1.2 The Paradip Refinery Project (PDRP), in Orissa State, India is classed (for the purpose of this project specification) as having a climate of 'severe tropical industrial/marine' atmosphere. As a result of the local environmental conditions, for painting purposes the site shall be considered as a "CM5" category as per ISO 12944-6. The overall aim of this specification is to achieve a life to first full maintenance of ten (10) years, not including regular maintenance touch ups/repairs.

1.1.3 The protective coating systems specified include surface preparation, paint material selection, paint application method, inspection requirement and acceptable minimum dried film thicknesses.

1.1.4 Plant items covered by this Specification include above ground piping, equipment, un-galvanised structural steelwork, furnace plating, storage tanks and other miscellaneous unprotected steel items.

1.1.5 This Specification excludes protective coatings applied specifically for transportation and shipment, and painted markings applied to identify materials of construction.

1.1.6 This Specification provides guidance on generic internal lining systems. All internal lining systems shall be verified as being suitable for service by each EPCM/BOO based on the process design conditions present. Any new internal lining systems that are proposed by the EPCM/BOO shall be submitted for MPMC review and shall include a basis for selection.

1.1.7 This Specification does not specify high temperature resistant painting systems for operating temperatures above 600°C, nor temperature indicating paints for refractory lined equipment items.

1.2 Reference Standards

1.2.1 The latest editions and revisions of the following Standards shall be used to supplement and define the intentions and requirements of this Specification.

British Standards

B.S. 5493	(Previously C.P. 2008) Code of Practice for Protective Coating of Iron and Steel Structures against Corrosion.
B.S. 7079	Preparation of steel substrates before application of paints and related products.
B.S. 1461	Hot Dip Galvanised Coatings on iron and Steel Articles.

Swedish Standard

SIS 05 59 00	Pictorial Surface Preparation Standards for Painting Steel Surfaces.

American Steel Structures Painting Council Standards

SSPC - PA1	Shop, Field and Maintenance Painting
SSPC - PA2	Measurement of Dry Paint Thickness with Magnetic Gauges
SSPC - SP1	Solvent Cleaning
SSPC - SP2	Hand Tool Cleaning
SSPC - SP3	Power Tool Cleaning
SSPC - SP5	White Metal Blast Cleaning
SSPC - SP6	Commercial Blast Cleaning
SSPC - SP7	Brush-off Blast Cleaning
SSPC - SP8	Pickling
SSPC - SP10	Near White Blast Cleaning

International Standards

ISO 12944-1	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 1: General introduction
ISO 12944-2	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 2: Classification of environments

ISO 12944-3	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 3: Design considerations
ISO 12944-4	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 4: Types of surface and surface preparation.
ISO 12944-5	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 5: Protective paint systems.
ISO 12944-6	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 6: Laboratory performance test methods.
ISO 12944-7	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 7: Execution and supervision of paint work.
ISO 12944-8	Paints and varnishes -- Corrosion protection of steel structures by protective paint systems -- Part 8: Development of specifications for new work and maintenance.
IISO 2811	Paints and varnishes - Determination of density
ISO 1515	Paints and varnishes - Determination of volatile and non-volatile matter.
ISO 2431	Paints and varnishes -Determination of flow time by use of flow cups
ISO 1524	Paints, varnishes and printing inks - Determination of fineness of grind
ISO 8501-1	Preparation of steel substrates before application of paints and related products — Visual assessment of surface cleanliness. Part 1: Rust grades and preparation grades of uncoated steel substrates and steel substrates after overall removal of previous coatings
ISO 8502-1	Preparation of steel substrates before application of paints and related products —Tests for the assessment of surface cleanliness-Part 1: Field test for soluble iron corrosion products

ISO 8502-2	Preparation of steel substrates before application of paints and related products - Tests for the assessment of surface cleanliness -- Part 2: Laboratory determination of chloride on cleaned surfaces
ISO 8502-3	Preparation of steel substrates before application of paints and related products — Tests for the assessment of surface cleanliness — Part 3, Assessment of dust on steel surfaces prepared for painting (pressure-sensitive tape method)
ISO 8502-9	Preparation of steel substrates before application of paints and related products -- Tests for the assessment of surface cleanliness -- Part 9: Field method for the conductometric determination of water-soluble salts
ISO 8503-2	Preparation of steel substrates before application of paints and related products — Surface roughness characteristics of blast cleaned steel blast-cleaned steel-Comparator procedure
ISO 8504-2	Preparation of steel substrates before application of paints and related products — Surface preparation methods — Part 2, Abrasive blast-cleaning
ISO 2808	Paints and Varnishes - Determination of film thickness
ISO 4624	Paints and Varnishes - Pull-off test for adhesion

American Society for Testing Materials

ASTM B117	Salt Spray (Fog) Testing
ASTM D4285	Standard Test Method for Indicating Oil or Water in Compressed Air

NACE

RP 0188	Discontinuity (Holiday) Testing Of Protective Coatings

Indian Codes & Standards

IS 14428	Guidelines for painting of structures in aggressive chemical environment

IS 9954	Pictorial Surface Preparation Standards for Painting of Steel Surfaces

Project Documents

PDRP-8440-SP-0008	Coating & Wrapping of Buried Pipework
PDRP-8440-SP-0007	Passive Fire Protection & External Fireproofing of Equipment Support and Structure

1.2.2 The latest edition and revision of British Standard 1461 shall be used to define minimum requirements for hot dipped molten zinc coatings of steel items designed to be galvanised.

1.2.3 Specific Project requirements shall be defined and specified which shall include National and Local Health and Safety requirement.

1.3 Definitions

1.3.1 'SHOP' shall be defined as preparing and painting of new, bare steel surfaces in a Manufacturer's or Fabricator's shop or plant works prior to transportation to the site of construction or a controlled environment paint facility erected at the permanent site of construction.

1.3.2 'FIELD' shall be defined as preparing and painting of new bare or shop primed steel surfaces or even completely painted shop steel surfaces at the permanent site of plant construction.

1.3.3 'T.U.' shall be defined as the painting of plant items that have previously been primed and/or overcoated and require a maintenance or a repair touch-up coating.

1.3.4 'PROTECTIVE PAINTING SYSTEM' shall be defined as the complete and total protection including surface preparation and intercoat cleaning, primer material, undercoat and top coat materials.

1.3.5 'D.F.T.' shall be defined as the minimum dried film thickness.

1.3.6 ‘CONTRACTOR’ shall be defined as the supplier of equipment and support services for a particular piece of equipment/package.

1.3.7 'APPLICATOR' shall be defined as the party that is responsible for surface preparation, priming, and application of painting materials in a CONTRACTOR'S facility, in a SHOP or in the FIELD.

1.3.8 'MAIN CONTRACTOR' shall be defined as the party responsible for the EPCM/BOO/BOOT scope of work.

1.3.9 'MANUFACTURER' shall be defined as the party supplying the coating material.

1.3.10 'IOCL' shall be defined as Indian Oil Corporation Limited (Client).

1.3.11 'EPCM' shall be defined as Engineering, Procurement and Construction Management Contractor.

1.3.12 'BOO' shall be defined as Build, Own, Operate Contractor.

1.3.13 'MPCM' shall be defined as Managing Project Management Consultant.

1.4 Design Requirements

1.4.1 All equipment shall be supplied by the CONTRACTOR to the job site with a full protective paint system as described herein. The surface preparation, painting application and inspection etc. shall be conducted under SHOP conditions in order that no further painting work shall be required at the job site. Equipment such as site erected tanks shall be permitted to be FIELD painted only when approved by the MAIN CONTRACTOR for practical reasons.

1.4.2 Shop applied coatings shall be dried and cured sufficiently to permit transport to the site of construction without excessive damage.

1.4.3 Field applied finishing coatings shall be subject to the requirements of this Specification and shall include repair and touch-up of shop applied coatings.

1.4.4 Shop and Field weld margins subject to Field testing shall be Shop surface prepared and taped for a distance of 50mm either side of the weld line, after successful primer application there shall be a further 50mm taped margin applied for each subsequent coat to aid in coating thickness build up. The

protective tape shall be sufficient quality to protect the achieved profile from deterioration. The remainder of the paint shall be chamfered back for approximately a further 100 mm to 150 mm.

1.4.5 The CONTRACTOR shall remove slag spatter and flux residues from welded areas and grind out any sharp edges. Sharp edges, fillets, corners, and weld seams shall be radiused or smoothed to a minimum of 3 mm. All bolt holes shall be drilled and ground prior to blast cleaning. When grinding, weld or base metals shall not be ground below the surface. Prior to painting the weld and surrounding area shall be prepared by washing thoroughly with fresh water followed by chipping or blast-cleaning. Preparation shall include the cleaning of the chamfered paint edges and the removal of any burnt or scorched paint.

1.4.6 Particular attention shall be paid to the preparation and painting of corners, edges, welds, rivet heads, small brackets, nuts and interstices. Prior to applying each full coat of the coating system, these positions shall be pre-coated by brush, i.e. 'stripe coated' (using a contrasting shade to the main coat where possible) to ensure specified minimum dried film thicknesses are achieved.

1.4.7 Care shall be taken to preserve the friction characteristics of faying surfaces of structural connections made with high strength friction - grip bolts and guidance given in B.S. 5493 Clause 25 shall be followed. However, as inorganic zinc primers offer a good slip coefficient, faying surfaces already primed with inorganic zinc shall only require surfaces to be cleaned and dried prior to bolting together. Zinc rich epoxy primers are not suitable for faying surfaces.

1.4.8 Aluminium paint shall not be applied to any steel or iron surfaces likely to attain a temperature greater than 100°C in any building where inflammable vapour may be present.

1.4.9 At the site of construction the mechanical contractor shall be responsible for properly handling and storing the primed and painted items prior to erection to minimise damage to primed and painted surfaces.

1.4.10 The protective coating system applied by the MANUFACTURER to proprietary items such as motors, pumps, turbines, instrument panels etc. shall be suitable for the intended operational and environmental conditions at the site of construction.

1.4.11 The selected protective paint systems take into consideration the corrosive environment, time delays between coating applications and specified maximum normal operational surface temperatures. Special consideration shall be given to items such as refractory lined piping and ducting, flare gas riser piping and protective heating tracing systems. In addition, anticipated conditions during start-up, shut-down and maintenance operations shall taken into account.

1.4.12 Structural supports such as Vessel Skirts, legs, saddles etc., shall be considered as operating below 93°C unless otherwise specified.

1.4.13 All un-insulated portions of insulated equipment comprising vessel nozzles, man-way covers, brackets, clips, valves, relief valves, etc. shall be painted in accordance with temperature conditions involved. All un-insulated portions of piping, including pipe supports etc. shall be painted in accordance with the maximum normal operational surface temperatures involved.

1.4.14 The selected protective paint systems shall generally be based on the maximum normal operating temperatures of pipework or equipment items with the following exceptions:-

a) items internally lined where paint systems shall be selected to suit the anticipated outside wall temperatures.

b) items subjected to pre-commissioning steam out where the paint system employed shall take into account the designated steam out temperature and shall be suitable for that temperature.

1.4.15 The following surfaces and items shall not be painted unless specifically instructed otherwise.

a) Hot dipped galvanised steel (unless for colour identification), plastic including plastic coated materials, aluminium and other project nominated non-ferrous metals.

b) Deleted.

c) Asbestos free cement sheeting, slate, brick, tile, glass and similar weather resistant materials.

d) Insulation finishing materials.

e) Concrete or Mortar fireproofing.

f) Machined surfaces of machinery and gasket contact surfaces.

g) Any equipment furnished completely primed and finish coated by the equipment manufacturer (pumps, motors and other machinery, instruments, instrument boards etc.)

h) Equipment name plates or any special instructions attached or included as part of an equipment unit.

i) Tripping mechanisms of Steam Turbines, Valve stems movable linage on equipment, grease fittings, compressor or pump shafts or any such surfaces on equipment that are normally lubricated or have close working tolerances.

1.4.16 The items listed below shall be shielded and protected to prevent damage during surface preparation and paint material application operations. All openings, including those that are flanged or threaded shall be sealed to prevent entry of blast abrasive or paint material. After completion of painting operations all material used for shielding and sealing shall be removed unless instructed otherwise.

a) Nameplates
b) Packing glands and seals
c) Bearings
d) Rotating equipment couplings and shafts
e) Lubrication fittings
f) Pressure gauges
g) Gauge and Flow Indicator glasses
h) Motor starters
I) Instrument dials
j) Vents
k) Exposed linkages
l) Valve stems
m) Light bulbs, enclosures and reflectors
n) Air intakes
p) Rubber parts and plastic

1.4.17 Surfaces requiring Personnel Protection shall be fully painted in a similar manner to an un-insulated surface.

2.0 SURFACE PREPARATION

2.1 Shop Preparation

2.1.1 Prior to commencement of surface preparation all surface defects likely to be detrimental to the protective painting system shall be removed. All fins at saw cuts, burrs and sharp edges shall be similarly removed. Where extensive grinding has been necessary the dressed areas shall be re-prepared to remove all rust and provide an adequate paint key.

2.1.2 Prior to blast cleaning all grease and oil contamination shall be removed by wiping or scrubbing the surface with rags or brushes wetted with solvent followed by a wipe down with clean dry cloths. Solvent cleaning shall be carried out in accordance with SSPC SP-1.

2.1.3 Blast cleaning shall not be carried out whilst the temperature of metal surfaces are less than $3^{o}C$ above the dew point of the surrounding air or when the relative humidity of the air is greater than 85%.

2.1.4 All blast cleaned surfaces shall be primed before visible re-rusting occurs or within four hours of the commencement of blasting, whichever is the sooner.

If visible re-rusting occurs then the metal surfaces shall be lightly reblasted and where necessary washed to remove rust blooming and accumulation of soluble salts. Spent blast abrasive shall be completely removed from the prepared surface by either vacuum cleaning or stiff-brush sweeping.

2.1.5 Soluble salt contamination tests shall be carried within 4 hours in accordance with ISO 8502-9. The maximum level of contamination shall be $5\mu g/cm^2$. Surfaces that do not meet this criteria shall be reprocessed.

2.1.6 When additional areas are blasted adjacent to newly primed areas, at least 50 mm of the previously primed area shall be reblasted, with the abrasive directed away from the coated surface.

2.1.7 The maximum and minimum vertical distance between the summit of any peak and the depth of an immediate adjacent trough on blast cleaned surfaces shall be in accordance with the MANUFACTURER's recommendations. In addition to visual examination, random measurements of blast / anchor profile shall be made at a minimum frequency of five (5) per shift per work area, or one per ten (10) square metres if greater, using Testex tape or other approved method. The surface profile measurements shall not be less than minimum value specified by the MANUFACTURER.

2.1.8 Regular measurement of the roughness of the blasted surface shall be carried out using one of the following methods:

1. The roughness of the blast cleaned surface is initially measured with a flat-ended calibrated micrometer. The surface is then ground until the bottoms of the deepest pits only are visible, and a further thickness measurement taken. (Suitable instruments are the Elcometer surface profile meter or Microtest).

2. The roughness of the blast cleaned surface is visually compared with a reference surface. Suitable comparators are Rugotest or Keane-Tator Surface profile comparator.

Only dry blast cleaning techniques shall be employed and abrasives shall be expendable copper slag or re-usable iron and steel grit. A non-metallic blast abrasive, i.e. aluminium oxide or garnet shall be used when abrading stainless steel surfaces.

The use of silica sand, potentially silica containing material, and copper slag abrasives are not permitted and shall not be used under any circumstances.

2.1.9 Abrasive size shall be selected to best suit the type of priming material and to satisfy the MANUFACTURER'S recommendation bearing in mind the restriction imposed by Paragraph 2.1.6 and 2.1.7 above.

2.1.10 At the commencement of each application, sample metal panels shall be prepared to demonstrate the ability to comply with the specified level of cleanliness and, after approval, shall be used as the standard for the remainder of project work.

2.1.11 Surface preparation of stainless steel shall be in accordance with ISO 8504-2, Sa 1 light blast cleaning to achieve a 25-40μm profile.

2.1.12 Defects such as holes, crevices, voids and gaps that are not structurally detrimental, shall be filled with solvent-less epoxy filler as agreed by IOCL. The filler shall be applied after abrasive blasting and prime coating but before finishing coating.

2.1.13 Abrasive material for blast cleaning, consisting solely of steel shot shall not be used. A mixture consisting of steel shot and at least 25% by weight steel grit is acceptable.

2.1.14 Spent blast abrasive shall be completely removed from the prepared surface by either vacuum cleaning, blowing down with clean dry compressed air, or stiff brush sweeping.

2.1.15 Expendable abrasive used for blast-cleaning shall be free of contaminants such as chlorides and other soluble salts and shall not contain metallic copper, nor more than 2% by weight of copper oxide. Expendable abrasives shall not be recycled.

2.1.16 The abrasive used in installations such as wheel abators or manual blast rooms/cubicles in which the abrasive is recovered and re-used shall be a mixture of chilled iron or steel grit and steel shot able to produce the required surface profile. The abrasive mixture shall be replenished using new and worn abrasive so as to produce a consistent profile height and standard of surface cleanliness. The abrasive mixture shall be kept free of dust (including metallic particles) and debris.

2.1.17 Abrasive blast rooms/cubicles shall be fitted with approved reclamation and cleaning systems.

2.1.18 Blast cleaning shall not be carried out in the vicinity of painting works and/or where it can result in the contamination of wet paint. Care shall be taken protect adjacent areas and equipment from blasting by the erection of screens and barriers.

2.1.19 Acid washes and other cleaning solutions or solvents including inhibitive washes intended to prevent rusting shall not be used on metal surfaces after they are blasted.

2.1.20 A safety checklist is provided in section 4.2

2.2 Field Preparation

2.2.1 All the provisions for SHOP Surface Preparation shall be applicable for FIELD Preparation. If the specified requirements are impractical because of FIELD Conditions, then the MAIN CONTRACTOR shall be consulted.

Any revisions to – or, deviations from – this specification shall only be acceptable when approved in writing by the MAIN CONTRACTOR on behalf

of IOCL. Written instructions shall be given prior to any amended coating related activities being executed.

2.2.2 Shop prepared and primed surfaces shall be Field repaired after erection of primed plant.

2.2.3 Over areas considered to be inaccessible after erection shop applied primer repairs, field applied primers, undercoats and finishing coats shall be applied, dried and cured before construction takes place. Care shall be taken to allow sufficient drying and curing time before moving, lifting or erecting newly painted surfaces.

2.2.4 Prior to mechanical preparation and application of priming paint, the metal surface and feathered area of shop coating shall be solvent cleaned and thoroughly washed with clean water to remove all soluble salts and dried in a manner which will not re-deposit contaminants on to the cleaned surface.

Field surface preparation local to weld areas shall be carried out by vacuum blast method as required in the painting system far as is practicable within the field construction conditions. Where surface cleaning by abrasion blasting is prohibited, weld and other small areas shall be prepared by use of Power Tool Cleaning or Hand Tool Cleaning or a combination of these methods depending on the contour and position of the surface to be painted and a Main Contractor approved primer material suitable for the cleaning method selected. Power Tool Cleaning or Hand Tool Cleaning shall only be carried out once approval has been provided by IOCL.

Any revisions to – or, deviations from – this specification shall only be acceptable when approved in writing by IOCL, written instruction shall be given prior to any amended coating related activities being executed.

2.2.5 Surface preparation with Power Tool or Hand Tools shall be pursued to achieve a minimum level of Cleanliness equivalent to the Swedish Pictorial Standard St 3.

2.2.6 The cleaning method(s) undertaken shall remove all weld flux slag and fume weld heat oxides and weld metal spatter in addition to all loose mill scale or loose paint if present after welding and testing has been completed.

2.2.7 The use of grinding discs may be included to achieve the removal of tightly adhering weld splatter, but great care shall be taken to ensure that surfaces shall not be polished nor left with a detrimental roughness or gouge marks.

2.2.8 The final action of surface preparation prior to the application of priming paint shall be a thorough washing with clean water to remove all soluble salts from the metal surface and a "drying" action which will not re-deposit contaminants on to the cleaned surfaces.

2.2.9 The initial coat of priming paint shall be carefully applied to ensure a thorough "wetting" of the prepared surfaces.

2.2.10 Shop applied paint shall be feathered back to allow the correct application and build up of required site applied paint coatings.

3.0 APPLICATION CONDITIONS

3.1 Weather and Surface Conditions

3.1.1 Surface preparation and painting shall not be undertaken in conditions which are favourable to surface condensation or outside daylight hours on exterior locations.

3.1.2 Conventional solvent based paints shall not be applied to exterior surfaces in damp, humid weather when air temperature is below 5°C or when the metal surface temperature is less than 3°C above the dew-point temperature.

3.1.3 Chemically cured paints shall not be applied at air or metal surface temperature below 10°C or when relative humidity is above 85%.

3.1.4 Before the application of paint materials, the APPLICATOR's paint procedure shall contain the MANUFACTURER's written recommendations regarding temperatures and ambient conditions of application.

3.1.5 Paint materials shall not be applied over metal surfaces having temperatures in excess of 45°C unless confirmed by MANUFACTURER datasheets and approved by IOCL.

3.1.6 When a metal surface to be coated exceeds 38°C, the use of sun screens/shades to maintain workable temperatures should be considered. Tarpaulins or monoflex sheeting type habitats could help. MANUFACTURER's written advice to be sought.

3.1.7 Paint materials shall not be applied to steel when temperatures are expected to fall to 0°C before the paint has cured.

3.1.8 Paint materials shall not be applied in rain, snow, fog or mist, nor to wet or damp surfaces. Paint shall not be applied to frost or ice coated surfaces.

3.2 Compressed Air

3.2.1 Compressed air supply used for blast cleaning shall be free from water and oil. Adequate separators and traps shall be provided and installed in the coolest part of the system; they shall be regularly emptied to prevent carry over of water and oil. Accumulations of oil and moisture shall be removed from the air receiver by regular purging. Air compressor shall not deliver air at a temperature in excess of 110°C.

The dryness and cleanliness of the compressed air supply used for blast cleaning shall be verified by testing the air on a white blotter as per ASTM D 4285. The testing shall be carried out at the start of each shift and every 4 hours thereafter and at the restart of compressor operation when off for more than 15 minutes.

Compressed air pressure shall be tested at the start of each shift and every 4 hours thereafter in order to ensure an even blast profile.

3.2.2 Nozzles shall be discarded and replaced when over worn.

3.2.3 Where air-operated equipment is used, the operator's hood or head gear shall be ventilated by clean cool air served through a regulator filter, to prevent blast cleaning residues from being inhaled.

3.3 Paint Mixing

3.3.1 Paint material shall be thoroughly mixed immediately prior to application. Mixing shall be by means of mechanical stirrers, paddle mixers, can vibrators or can shakers.

3.3.2 Constant mechanical agitation shall be maintained in the paint reservoir throughout paint spraying operations.

3.3.3 Thinners shall only be added strictly in accordance with the MANUFACTURER's written instructions.

3.3.4 The type of thinner shall comply with the MANUFACTURER's written instructions.

3.3.5 When thinners are permissible, they shall be added during the paint mixing process in accordance with the MANUFACTURER's datasheets.

3.4 Film Thickness

3.4.1 The applied DFT of each coat and the total system shall not be less than the specified minimum stated in section 9.2

3.4.2 The DFT shall be measured in accordance with SSPC – PA2. The average of any five (5) spot readings on each $10m^2$ area shall be equal to or greater than the specified minimum DFT and no single point reading shall be less than 80% of the specified minimum DFT. Each spot reading shall be made up of average of three (3) gauge readings that shall be taken in a 40mm diameter circle defining the spot.

3.4.3 DFT measurement shall be made of each coat and of the total system.

3.4.4 The total minimum dry film thickness for a multi-coat system shall not be less than the added total specified minimum dry film thicknesses.

3.4.5 The dry film thickness of individual coats or system shall not exceed the MANUFACTURERs recommended maximum thickness, and shall not prove detrimental, i.e. wrinkling, mud cracking, general softness, poor adhesion, etc., to both the coating and/or the total system.

3.4.6 Inadequate film thickness shall require the further application of an additional complete coat over the whole area until the dry film thickness is sufficient to meet the specified minimum. Excessive thick coats shall be removed completely by blast cleaning, and shall be re-coated according to this specification.

3.4.7 Film thickness measuring shall not include thickness of pre-fabricating primers. In the absence of pre-fabricating primer film thickness records, 25 microns shall be allowed for pre-fabrication primers whilst checking further film coat thicknesses.

3.5 Application

3.5.1 Paint materials shall only be applied by the MANUFACTURER's recommended method in accordance with the paint datasheet.

3.5.2 Spray guns shall be held perpendicular to the surface being coated at a maximum distance of 300 mm from the surface and no closer than 200 mm to the surface. Wet film thickness (comb) gauges shall be used by paint spray gun operatives to ensure an even and adequate DFT will be achieved.

3.5.3 Material and air pressure on the spray guns shall be regulated to achieve optimum atomization of the paint with the lowest possible pressures to avoid spray ricochet from the surface being painted.

3.5.4 Overlapping horizontal passes shall be made for each spray coat to obtain uniform application. Particular care shall be exercised to ensure that a uniform, complete wet coat is applied. Semi-dry spraying results in poor flow-out with excessive pinholing and shall be avoided.

3.5.5 Coats of paint shall be uniformly applied without sags, runs, spots, damage or contamination; such irregularities shall be removed and the affected area recoated.

3.5.6 Each coat of paint shall be in a proper state of cure or dryness before the application of the succeeding coat. Attention shall be paid to the MANUFACTURER's datasheet for curing times between coats.

3.5.7 Equipment and/or piping shall not be painted (with the exception of IOCL approved pre-fabrication primer) until all approved heat treatment, non-destructive testing, and pressure testing has been completed. Tank bottom coating systems are an exception to this requirement.

3.5.8 Damage to each coat of paint, whether shop or field applied, shall be repaired by abrasive paper cleaning and spot priming before succeeding coats of paint are applied. No succeeding coat of paint shall be applied before preceding coat of paint is thoroughly dry.

3.5.9 SHOP primed surfaces contaminated during transport and site storage shall be thoroughly washed down with fresh clean water (<30 ppm chlorides) and allowed to dry before the application of further paint coatings. A soluble salt

contamination test shall be carried out prior to paint application in accordance with 2.1.5.

3.5.10 Coatings which have weathered for more than six (6) months shall be thoroughly inspected prior to application of further coats to prevent intercoat adhesion problems.

3.5.11 To minimise intercoat contamination, succeeding coats shall be applied with a minimum time lapse consistent with proper drying time, work and erection schedules in accordance with the MANUFACTURER's datasheet.

3.5.12 Roller application of the first primer coat is not acceptable. Rollers when used shall be manufactured from good quality 'carpet pile' or lamb's wool.

3.5.13 When paints are applied by brushing, brushes shall be of a style and quality that will permit proper application of the material. Brushing shall be done so that a smooth coat as nearly uniform in thickness as possible is obtained. There shall be no deep "detrimental" brush marks.

3.6 Repairs

3.6.1 SHOP prepared and primed surfaces shall normally be FIELD repaired after erection of primed plant, unless covered by 3.6.2.

3.6.2 Over areas considered to be inaccessible after erection SHOP applied primer repairs, FIELD applied primers, undercoats and finishing coats shall be applied, dried and cured before construction takes place. Care shall be taken to allow sufficient drying and curing time before moving, lifting or erecting newly painted surfaces.

3.6.3 Damage to each coat of paint, whether SHOP or FIELD applied, shall be repaired by abrasive paper cleaning and spot priming before succeeding coats of paint are applied. No succeeding coat of paint shall be applied before preceding coat of paint is thoroughly dry.

4.0 APPLICATION RESPONSIBILITY

4.1 General

4.1.1 Certificates of release, confirming that successful completion of all appropriate tests and safety checks have been carried out to the released working areas, shall be obtained before the commencement of painting work.

4.1.2 All provisions of this Specification shall apply to both SHOP and FIELD painting.

4.1.3 All paint work shall be carried out in a neat, thorough and workmanlike manner.

4.1.4 Wet paint shall be protected against damage by the APPLICATOR.

4.1.5 All shipping tags, wires, strings and other means of temporary or shipping identification on surfaces to be painted shall be removed only AFTER checking with and receiving authorisation from the Main Contractor's Representative.

4.1.6 The APPLICATOR shall be responsible for protection against damage to abutting property, vehicles, pedestrians and other portions of the structure, due to the painting operations.

4.1.7 All instruments, gauge glasses, etc. (see 1.4.15) be protected from paint splatter by the painting contractor.

4.1.8 The job site shall be maintained clean and neat during the paint application work.

4.1.9 Paints, thinners, solvents and rags, waste etc., soiled with these materials shall be kept in tightly closed containers whilst on the job site and not in use.

4.1.10 The MAIN CONTRACTOR's Representative or his designate shall have the right to stop the application of paint when in his opinion; conditions may result in injury to the applied coatings.

4.1.11 The APPLICATOR shall provide a temporary shelter for storage and mixing of paint materials.

4.1.12 During both application and drying, adequate ventilation shall be provided if the work area is enclosed.

4.2 Safety Requirements

4.2.1 The paint contractor shall be acquainted with safety and security requirements for equipment and materials used.

4.2.2 The paint contractor shall observe and carry out all work in strict compliance with plant, local and national Health and Safety rules and shall obtain all necessary permits from the applicable authorities.

4.2.3 The paint contractor shall be responsible for earthing all his mechanical equipment to prevent build-up of static electricity.

4.2.4 Before painters, blasters or general operatives are sent to work in any plant area; the working environment should be studied to detect hazards such as concentration of noxious fumes, presence of splash or spillage of harmful liquids, conditions of excessive heat, or areas where oxygen content of the air may be dangerously low. Where these conditions exist, they must be corrected before painting personnel are permitted to enter the area, or painting personnel must be protected by materials or devices that will allow them to work without fear of injury or illness that might otherwise result from exposure.

4.2.5 The following is a checklist for memory jogging purposes only, it is not a comprehensive list, and it does not alleviate any responsibility of the paint contractor to ensure the welfare of his workforce and any other workers in the surrounding area:

Blasting and Coating Operations – Safety Checklist

Abrasive Blasting	**Confirmation or Comments**
Written release for painting works	
Potential for trips, slips or falls	

Abrasive Blasting	**Confirmation or Comments**
Barricade around work area and warning notices placed in conspicuous locations	
Well trained operators	
Air fed helmet and air purifier (in good working order)	
Good coveralls, cape and gauntlets	
Appropriate safety boots	
Ear protectors (where needed)	
Dead-mans handle for operator	
Antistatic hoses with external quick fittings	
Intercom or clear sign language between the blaster and the pot man	
All compressors, hoses, nozzles and ancillary items are in good efficient working order	
Painting	
Well trained operators	
Safety boots	
Overalls	
Gloves	
Eye protection	
Ear protectors (where needed)	
All tools and equipment are appropriately calibrated and are in good working order	

4.2.6 The quantity of Volatile Organic Compound (VOC), in grammes per litre, (pounds per gallon), shall be in compliance with the legal local, national and international values.

4.3 Inspection Requirements

4.3.1 All phases of cleaning and painting and galvanising operations of all shop and field painting shall be subject to spot check inspection by the Main Contractor's and IOCL appointed Representative.

4.3.2 This periodic inspection will be made to ensure that the APPLICATOR is accurately following the recommendations of this specification and the MANUFACTURER's instructions.

4.3.3 The paint applicator is required to keep an accurate daily record of air temperature and humidity conditions and the times of commencement and cessation of all phases of the cleaning and painting operations. These records which shall be certified accurate by the paint applicators' painting supervisor shall be available for inspection by the MAIN CONTRACTOR at all times.

4.3.4 The responsibility for the total cleaning, painting and galvanising operations remain entirely with the applicators who shall fully guarantee material and workmanship for the entire protective system. Inspection, approval or comments by THE MAIN CONTRACTOR, their agents or by independent inspectors shall not relieve the applicator of this responsibility.

4.3.5 The MANUFACTURER/CONTRACTOR shall provide a FIELD supervisor during the painting work. The supervisor shall hold a minimum qualification of NACE CIP Level 2 or ICorr/Frosio equivalent.

4.3.6 It is the coating APPLICATOR's responsibility to ensure that sufficient tests are carried out to show that the coatings comply with this specification. COMPANY reserves the right to carry out any additional tests at random to ensure compliance and the coating APPLICATOR shall have suitable test-equipment available for COMPANY use.

4.3.7 Prior to acceptance, COMPANY will carry out a final inspection. Representatives of the APPLICATOR and MANUFACTURER may be present. The coating APPLICATOR shall repair any defective areas at no additional cost to IOCL.

4.3.8 Acceptable finish work must be free of abrasions and must be uniform in colour and appearance.

4.4 Daily Inspection and Reporting

4.4.1 The APPLICATOR shall produce a daily report recording all relevant information appertaining to the surface preparation and coating activities. The report shall contain details on the items being coated, coating materials consumed by product number, description and batch number. The report shall include the names of operators who carried out blast cleaning and coating application, the time started and finished and the number of litres consumed. In the event of a coatings failure, it should be possible to retrieve the relevant surface preparation and coatings data for the item whose internal lining system has failed.

4.4.2 The coating APPLICATOR shall also keep a daily log of environmental conditions as follows:

- Air and steel temperatures
- Humidity

The above readings shall be recorded as follows:

- During blasting at each shift change
- During cleaning every four hours
- During application every shift change
- During drying/curing every shift change

4.5 Inspection Tasks and Criteria

4.5.1 All blast-cleaned internal surfaces shall be visually inspected to check for conformance to the standard of cleanliness, profile and roughness of the substrate as specified in this specification immediately prior to painting.

4.5.2 All painted surfaces shall be visually examined after application of each coat for deleterious film defects e.g. blisters, pinholes, dry spray, runs, sagging, wrinkling, mud-cracking, delamination, loss of adhesion, embedded particles, overspray and other contamination.

4.5.3 Wet film thickness measurements shall be made throughout the course of painting in order to ascertain the adequacy and uniformity of thickness. The readings shall be taken at least one every ten (10) square metres.

4.5.4 Dry film thickness (DFT) readings shall be taken using a non-destructive dry film thickness instrument capable of storing the readings. Sufficient readings shall be taken covering each coat prior to application of the following coat to ensure the correct required DFT.

4.5.5 The minimum measurement frequency shall be one (1) per two (2) square metres of item surface, with additional measurements taken at changes in section, corners or edges or a minimum off 5 readings per item whichever is the greatest. The recorded print-out of dry film thicknesses shall be retained with the inspection records together with marked up layout drawings clearly identifying the areas surveyed for future reference.

4.5.6 Defective work shall be corrected at the APPLICATOR's cost.

4.5.7 Frequency of inspection and criteria for acceptance of painting work shall be in accordance with the following table:

Test	Standard	Frequency	Acceptance Criteria
Surface cleanliness	ISO 8501-1	one per ten (10) square metres	Sa 2 ½
Surface profile	ISO 8503-2	one per ten (10) square metres	As per coating procedure/datasheet
Soluble salt contamination	ISO 8502-2	one per ten (10) square metres	<5µg/cm2
Dust	ISO 8502-3	one per ten (10) square metres	Rating 2 or better
Adhesion	ISO 4624	one per test panel, per shift	5 MPa
Holiday Detection	NACE RP0188	100% of surface	Zero defects
Dry Film Thickness	SSPC PA-2	one per two (2) square metres	Maximum and minimum dry film thickness not to be exceeded
Wet film Thickness	ISO 2802	one per ten (10) square metres	Consistent with providing required DFT.

4.5.8 Prior to final acceptance of the paint work, an inspection shall be made. The CONTRACTOR and IOCL shall both be represented and they shall sign an inspection report agreed upon by all parties. The report shall consist, as a minimum, of the following:

- Names of the APPLICATOR and the responsible personnel.
- Dates when work was carried out.
- Equipment and techniques used.
- Type and calibration of instruments used.
- Weather and ambient conditions.

- Painting periods.
- Condition of surface before preparation.
- Tools and methods used to prepare surface.
- Condition after preparation.
- Information on systems being applied.
- Mixing and testing prior to application.
- Paint application techniques.

5.0 GALVANISED ITEMS

5.1 Requirements

5.1.1 Galvanised items shall be coated using the hot dipped process in accordance with the latest revision of BS 1461.

5.1.2 Articles specified to be galvanised shall be completely fabricated prior to the application of the hot dip galvanising process.

5.1.3 Items shall not be hot dipped before surface preparation has been carried out in a full and correct manner.

5.1.4 Surfaces shall be free of old paint, oil, grease, weld, slag deposits and laminations; and rolling fabrication defects eliminated prior to hot dipping.

5.1.5 Coating weights shall comply with Table 1 of BS 1461.

5.2 Repairs

5.2.1 For touch-up of drilled and/or cut edge damage areas of galvanised steel, surfaces shall be hand tool cleaned and painted with one coat of zinc-rich epoxy to a nominal DFT of 60 micron. Major damage shall be re-galvanised in accordance with the above mentioned standard.

6.0 EQUIPMENT MANUFACTURERS STANDARDS

6.1 General

6.1.1 On standard production equipment e.g. Compressors coated with the Equipment Manufacturer's complete protective coating system it shall be the Manufacturer's responsibility to ensure that their specification coating system is capable of protection within the anticipated service operating temperatures and environment, without break-down or failure. The Equipment Manufacturer's coating system shall be suitable for a CM5 environment defined as per ISO 12944-6 and shall comply with this specification unless they are able to demonstrate they are able to provide a coating system equal to, or better than that stated herein.

The protective coating system shall also be able to survive handling, transportation and storage prior to erection without incurring major damage.

6.1.2 The Equipment Manufacturer shall submit his surface preparation and protective coating system to the Main Contractor/IOCL for approval prior to him carrying out any coatings related works.

7.0 MATERIALS

7.1 General

7.1.1 Coating materials shall be obtained from the following, alphabetically listed approved MANUFACTURER's using the full generic description given in **Section 9** of this specification:

- Asian Paints
- Carboline
- International Paints
- Jotun
- Kirloskar Corrocoat
- PPG (Sigma)

7.1.2 All material shall be supplied in the MANUFACTURER's original cans, durably and legibly marked with the description of the contents. This shall include the specification number, the colour reference number, the method of application for which it is intended, the batch number, date of manufacture and the MANUFACTURER's name, initials, or recognised trade mark. No intermixing of different brands or types of paints will be permitted.

7.1.3 The MANUFACTURER's shall supply data sheets for each type of paint which shall include the recommended minimum and maximum overcoating periods for the dry film thickness specified in this Specification.

7.1.4 Paint Systems, i.e. primer, undercoat, finish coat, shall be from the same MANUFACTURER to ensure compatibility.

7.1.5 All coating materials used shall conform to the following generalised composition clauses. In meeting the composition clauses, the MANUFACTURER shall provide evidence of compliance and shall produce premium quality paint with the full performance required by this specification:

- Zinc rich primers shall be based on two pack epoxy media and shall contain a minimum of 85% metallic zinc by weight in the dry film.
- Zinc phosphate two-pack epoxy primer shall be based on polyamide cured two-pack epoxy media and shall contain a minimum of 20% zinc phosphate by weight in the dry film.
- Zinc silicate primer shall be based on inorganic or ethyl silicate media and shall contain not less than 85% metallic zinc by weight in the dry film.
- Micaceous iron oxide pigmented paint shall contain a minimum of 50% natural laminar micaceous iron oxide by weight in the dry film.
- Glass Flake Epoxies paint shall contain a minimum of 20% glass flake by weight in the dry film. The applied painting system shall be compatible with cathodic protection systems and shall demonstrate excellent cathodic disbondment characteristics. When tested in accordance with ASTM G8 or similar, the maximum disbondment shall be less than 5mm.

7.1.6 Two-pack epoxy finishing paints shall be based on polyamide cured two-pack epoxy media and shall be pigmented with titanium dioxide and/or light fast coloured pigments to provide the necessary opacity, film build and weather resistance.

7.1.7 Aluminium silicone paint shall be based on a one-pack formulation comprising leafing aluminium pigment dispersed in silicone or modified silicone media. The paint shall be capable of withstanding continuous exposure to temperatures between 120°C and 500°C. Only ambient air curing heat resistant aluminium paints shall be applied (post heat curing materials are not acceptable).

7.1.8 Materials containing the following components shall not be used:

- Cadmium, lead or any toxic material to environment/personnel shall not be used.

- Paints for use over stainless steels shall not contain free chlorides or other halides and zinc. Chlorides or other halides tied up within the cured resin's molecule may be acceptable, unless they are subject to release through aging within the temperature range specified.

7.2 Material Storage

7.2.1 Storage of paint materials at the site of usage shall be the responsibility of the APPLICATOR and the storage planning shall take into account the geographical and meteorological conditions and make provisions for any abnormality such as temperatures. Paint shall be stored in dry, well ventilated, temperature controlled buildings out of direct sunlight in accordance with the MANUFACTURER's recommendations. Paint shall be used on a first in, first out basis and paint that has exceeded its shelf life shall not be used.

7.2.2 As a minimum all tins, bags, buckets, pails or other material storage containers for paints, thinners, abrasives or ancillary materials left on the job face shall be stacked in a safe, neat even manner off the ground and protected from the elements using a weather impervious sheet such as a tarpaulin. Fork lift truck pallets are generally acceptable for temporary stacking purposes. All items of material shall be stacked and marked indelibly for ease of identification.

7.2.3 The MANUFACTURER of the specified paint materials shall advise the APPLICATOR of the intended storage conditions and recommendations.

7.2.4 Records of paint material batch numbers together with dates of receipts of each batch shall be kept by the paint material applicator so that a sequence of storage can be arranged to ensure material is issued from store in the same sequence as received at painting venue. The Buyer reserves the right to inspect these records and check the issuing organisation at any time during the execution of contract.

7.2.5 Storage shall take into account and make provision for governing local safety regulations, adequate fire prevention equipment and its maintenance and the elimination of ignition risks (e.g. No smoking, use of flame proof equipment etc).

7.2.6 Purposeful spillage shall be prohibited, but sufficient inert absorbing material shall always be available for the immediate clean-up of accidental spillage.

The use of rags and wood dust for clean-up of inflammable materials is not permitted.

7.2.7 The disposal of waste and empty containers shall be subject to regulations laid down before the commencement of material deliveries. Empty and/or partially empty containers shall be collected and disposed of in accordance with the regulations.

7.2.8 Paint materials are potentially harmful to operatives and it shall be necessary to maintain a high standard of personnel hygiene. Paint Contractors shall advise the Main Contractor of details of precautions being taken to protect all working personnel in the vicinity of paint storage and handling.

8.0 FINISH COLOUR SCHEDULE

8.1 General

8.1.1 This section covers the general colour requirements for structural steelwork and equipment with operating temperatures below 120ºC:

Structural Steel, Tanks, Spheres, Electrical Equipment		
Sl. No:	**Service**	**Shade – Shade Number**
N/A	Structural Steelwork	Light Grey – RAL 7035
N/A	Spheres and Storage tanks	White – RAL 9003
N/A	Electrical Switch Boards, Cable Conduit and Transformers and all other electrical equipment	Manufacturers Painting standard (including RAL 7035) conforming to relevant code and practices prevailing in the country of manufacture
Un-insulated Equipment, Tanks and Structures		
Sl. No:	**Service**	**Shade – Shade Number**
-	Loading Arms (i) Structural Steel (ii) Arms	 Light Grey – RAL 7035 Yellow – RAL 1023
96.	Heater Structure	Signal Grey – RAL 7004
97.	Heater Casing	Aluminium – RAL 9006
98.	Vessels and Columns	Aluminium – RAL 9006
99.	Hydrogen Bullets	Antique Pink – RAL 3014
100.	LPG Vessels	Oxide Red – RAL 3009
101.	SO_2 Vessel	Yellow – RAL 1023
102.	Heat Exchangers	Aluminium – RAL 9006
103.	FO Tank and Hot Tanks	Black – RAL 9017
104.	All Other Tanks	Aluminium – RAL 9006
105.	Caustic/Amine/Acid Tanks	Gold/Yellow – RAL 1004
106.	Sour Water	Sky Blue – RAL 5015
107.	Outer Surface in Boiler House	Aluminium – RAL 9006
108	Steam Turbine	Aluminium – RAL 9006
109.	Compressors and Blowers	Dk Grey BS4800 18 B 25
110.	Pumps	Cobalt Blue RAL – 5013
111.	Motors (Except Fire Motors)	Bluish Green RAL 5021
112.	Hand Railing	Red – RAL 3001
113.	Staircase, Ladders and Walkways	Black – RAL 9017
114.	Load lifting equipment & mono rails etc.	Brown – RAL 8003

	Safety Colour Schemes	
Sl. No:	**Service**	**Shade – Shade Number**
115.	General Structure	Black – RAL 9017
116.	Switchgear (including inside sub-station)	Light Grey RAL 7035
117.	Dangerous Obstruction	Alternate Black (RAL 9017) and Orange (RAL 2008) Diagonal Banding
118.	Dangerous or Exposed Parts of Machinery	Orange – RAL 2008

8.2 Pipe Colour Bands

8.2.1 This section covers the requirements for a colour scheme identifying the contents of piping carrying products throughout IOCL refineries and petrochemical installations. The colour coding system is based on international specifications such as ASME, ANSI, BS and Indian Standards including IOCL's existing specification for colour coding.

8.2.2 The system of coding consists of a ground/base colour superimposed with secondary colour bands. The ground colour identifies the basic nature of the service and the secondary colour bands distinguish the particular service product contained. Ground colour shall be applied to the entire length of un-insulated piping.

8.2.3 The ground colours and secondary banding colours are defined in section 8.4.

8.2.4 The frequency of banding on un-insulated pipe shall be as follows:

- Unit Area – Bands at intervals of 6 metres
- Offsite Area – Bands at intervals of 10 metres

Each pipe segment will have a minimum of 1 identification band irrespective of length

8.2.5 Colour bands of the correct size shall be applied to the pipe, at:

- Both sides of valves, tees and other fittings
- Where pipes enter and emerge through walls

- Where pipes enter and emerge from walkway overpasses and battery limits.
- At uniform intervals along long sections of pipe
- Adjacent to tanks, vessels and pumps.

8.2.6 Insulated piping shall receive ground colouring and coloured (secondary) identification bands at a minimum of either side of valves, flanges and the like, at each change in flow direction and at no greater than 6 metre intervals, ground colours should be 2 metres long.

8.2.7 Secondary colours are to be applied using adhesive plastic tapes to the specified colour.

8.2.8 Bands widths are shown below for different pipe diameters and are to be spaced 25mm apart when two bands (or more) are used:

Outside diameter of pipe or covering (inches)	Width of colour bands (mm)
< 2”	200
2” – 4”	300
6” – 8”	600
≥ 10”	800

8.2.9 Bands shall also be displayed conspicuously near walkways, both sides of culverts, tanks, dykes, vessels, suction and discharge of pumps/compressors, unit battery limit, near valves of line, etc.

8.3 Identification Lettering

8.3.1 Name of service and direction of flow, for all lines shall be positioned at the following locations:

- Offsite lines: Both sides of culverts, any one side of walkways, near tank dykes, at tank inlet/outlet points and suction/discharge pumps/compressors.
- Unit lines: At the battery limit, suction/discharge of pumps/compressors, near vessels, columns, tanks, exchangers etc.

8.3.2 Identification/legend letter sizes on piping shall depend on the pipe diameter. Either white or black letters are to be selected so as to afford maximum contrast with the identification band colour.

Outside diameter of pipe or covering (inches)	Size of legend letters (mm)
< 2”	19
2” – 4”	32
6” – 8”	64
≥ 10”	89

8.3.3 Pipe contents and direction of flow is to be identified using legend letters and arrows, any hazard must be identified clearly by the legend.

8.3.4 Size of letters (stencilled or pre-formed adhesive) for equipment shall be:

Equipment	Size of legend letters
Column and vessel	150 mm (height)
Pump, compressor and other machinery	50 mm (height)

8.3.5 Lettering shall be black on pipes painted with light shade colours and white on pipes painted with dark shade colours to give good contrast

8.4 IOCL Paint Colour Code and Banding

8.4.1 The following base/ground and secondary colour designation for identification of various important services shall be followed:

Sl. No:	Service	Base/Ground Colour	Secondary Band Colours
	Hydrocarbon Lines – (Un-insulated)		
1.	Crude Sour	Dk Grey BS4800 18 B 25	(x1) Orange – RAL 2008
2.	Crude Sweet	Dk Grey BS4800 18 B 25	(x1) Red – RAL 3001
3.	Lube Oils	Dk Grey BS4800 18 B 25	(x1) Green – RAL 6002
4.	Flare Line	Aluminium – RAL 9006	Aluminium – RAL 9006
5.	L.P.G.	Orange – RAL 2008	(x1) Oxide Red – RAL 3009
6.	Propylene	Orange – RAL 2008	(x2) Oxford Blue – RAL 5003
7.	Naptha	Orange – RAL 2008	(x1) Green – RAL 6002
8.	M.S.	Orange – RAL 2008	(x1) Dk Grey BS4800 18 B 25
9.	AV. Gasoline (96 RON)	Orange – RAL 2008	(x1) Green – RAL 6002 + (x1) White – RAL 9003 + (x1) Red – RAL 3001
10.	Gasoline (regular, leaded)	Orange – RAL 2008	(x1) Black – RAL 9017
11.	Gasoline (premium, leaded)	Orange – RAL 2008	(x1) Blue – RAL 5017
12.	Gasoline (white)	Orange – RAL 2008	(x1) White – RAL 9003
13.	Gasoline (aviation 100/130)	Orange – RAL 2008	(x1) Red – RAL 3001
14.	Gasoline (aviation 115/145)	Orange – RAL 2008	(x1) Purple – RAL 4006
15.	N-Pentane	Orange – RAL 2008	(x2) Blue – RAL 5017
16.	Diesel Oil (white)	Oxide Red – RAL 3009	(x1) White – RAL 9003
17.	Diesel Oil (black)	Oxide Red – RAL 3009	(x1) Yellow – RAL 1023
18.	Kerosene	Oxide Red – RAL 3009	(x1) Green – RAL 6002
19.	HY. Kero	Oxide Red – RAL 3009	(x2) Green – RAL 6002
20.	Disulfide Oil (Ex-Merox)	Oxide Red – RAL 3009	(x1) Black – RAL 9017
21.	M.T.O.	Oxide Red – RAL 3009	(x3) Green – RAL 6002
22.	DHPPA	Oxide Red – RAL 3009	(x2) White – RAL 9003
23.	Flushing Oil	Oxide Red – RAL 3009	(x2) Black – RAL 9017
24.	Lab FS	Oxide Red – RAL 3009	(x2) Dk Grey BS4800 18 B 25
25.	Lab RS	Oxide Red – RAL 3009	(x3) Dk Grey BS4800 18 B 25
26.	Lab (Off. Spec.)	Oxide Red – RAL 3009	(x1) Lt Grey RAL 7036
27.	N-Paraffin	Oxide Red – RAL 3009	(x1) Blue – RAL 5017
28.	Heavy Alkylate	Oxide Red – RAL 3009	(x1) Red – RAL 3001
29.	Blow Down, Vapour Line	Aluminium – RAL 9006	(x1) Brown – RAL 8003

Sl. No:	Service	Base/Ground Colour	Secondary Band Colours
30.	Blow Down	Aluminium – RAL 9006	(x2) Brown – RAL 8003
31.	A.T.F.	Brown – RAL 8003	(x1) White – RAL 9003
32.	Toluene	Brown – RAL 8003	(x1) Yellow – RAL 1023
33.	Benzene	Brown – RAL 8003	(x1) Green – RAL 6002
34.	Lab Product	Brown – RAL 8003	(x1) Blue – RAL 5017
35.	Fuel Oil	Black – RAL 9017	(x1) Yellow – RAL 1023
36.	Fuel Oil (aromatic rich)	Black – RAL 9017	(x2) Yellow – RAL 1023
37.	Asphalt	Black – RAL 9017	(x1) White – RAL 9003
38.	Slop and Waste Oils	Black – RAL 9017	(x1) Orange – RAL 2008
39.	Slop Aromatic	Black – RAL 9017	(x2) Orange – RAL 2008

	Chemical Lines (Un-insulated)		
40.	Tri-Sodium Phosphate	Yellow – RAL 1023	(x1) Violet – RAL 4008
41.	Caustic Soda	Yellow – RAL 1023	(x1) Black – RAL 9017
42.	Sodium Chloride	Yellow – RAL 1023	(x1) White – RAL 9003
43.	Ammonia	Yellow – RAL 1023	(x1) Blue – RAL 5017
44.	Corrosion Inhibitor	Yellow – RAL 1023	(x1) Aluminium – RAL 9006
45.	Hexameta Phosphate	Yellow – RAL 1023	(x2) Black – RAL 9017
46.	Acid Lines	Gold/Yellow – RAL 1004	(x1) Red – RAL 3001
47.	Rich Amine	Yellow – RAL 1023	(x2) Blue – RAL 5017
48.	Lean Amine	Yellow – RAL 1023	(x3) Blue – RAL 5017
49.	Solvent	Yellow – RAL 1023	(x1) Green – RAL 6002
50.	LCS	Yellow – RAL 1023	(x1) Grey – RAL 7001

	Water Lines (Un-insulated)		
51.	Raw Water	Sky Blue – RAL 5015	(x1) Black – RAL 9017
52.	Industrial Water	Sky Blue – RAL 5015	(x2) Red – RAL 3001
53.	Treated Water	Sky Blue – RAL 5015	(x1) Oxide Red – RAL 3009
54.	Drinking Water	Sky Blue – RAL 5015	(x1) Green – RAL 6002
55.	Cooling Water	Sky Blue – RAL 5015	(x1) Pale Brown – RAL 8025
56.	Service Water	Sky Blue – RAL 5015	(x1) Red – RAL 3001
57.	Tempered Water	Sky Blue – RAL 5015	(x2) Green – RAL 6002
58.	DM Water	Sky Blue – RAL 5015	(x1) Aluminium – RAL 9006
59.	DM Water above 150ºF	Sky Blue – RAL 5015	(x2) Black – RAL 9017
60.	Sour Water	Sky Blue – RAL 5015	(x2) Yellow – RAL 1023

Sl. No:	Service	Base/Ground Colour	Secondary Band Colours
61.	Stripped Water	Sky Blue – RAL 5015	(x2) Blue – RAL 5017
62.	ETP Treated Water	Sky Blue – RAL 5015	(x2) Oxide Red – RAL 3009

	Fire Protection System (Above Ground)		
63.	Fire Water, Foam and Extinguishers	Red – RAL 3001	Red – RAL 3001

	Air and Other Gas Lines (Un-insulated)		
64.	Service Air	May Green – RAL 6017	(x1) Red – RAL 3001
65.	Instrument Air	May Green – RAL 6017	(x1) Black – RAL 9017
66.	Nitrogen	May Green – RAL 6017	(x1) Orange – RAL 2008
67.	Freon	May Green – RAL 6017	(x1) Yellow – RAL 1023
68.	Chlorine	Yellow – RAL 1023	(x1) Oxide Red – RAL 3009
69.	SO_2	Yellow – RAL 1023	(x2) White – RAL 9003
70.	H_2S	Orange – RAL 2008	(x2) Oxide Red – RAL 3009
71.	Gas (fuel)	Orange – RAL 2008	(x1) Aluminium – RAL 9006
72.	Gas (sour)	Orange – RAL 2008	(x2) Aluminium – RAL 9006
73.	Gas (sweet)	Orange – RAL 2008	(x1) Red – RAL 3001
74.	Hydrogen	Orange – RAL 2008	(x1) May Green – RAL 6017

	Steam and Condensate Lines (Un-insulated)		
75.	HP Steam &VHP Steam Line	Aluminium – RAL 9006	(x1) Yellow – RAL 1023
76.	MP Steam	Aluminium – RAL 9006	(x1) Red – RAL 3001
77.	MLP Steam	Aluminium – RAL 9006	(x1) Orange – RAL 2008
78.	LP Steam	Aluminium – RAL 9006	(x1) Green – RAL 6002
79.	Condensate	Sky Blue – RAL 5015	(x1) White – RAL 9003
80.	Condensate above 150ºF	Sky Blue – RAL 5015	(x3) Oxide Red – RAL 3009
81.	BFW	Sky Blue – RAL 5015	(x2) Traffic Red – RAL 3020
	Note: For all insulated steam lines, the colour coding shall be followed as given for un-insulated lines with the specified length of colour bands		

	Insulated Hydrocarbon Lines		
82.	IFO Supply	Black – RAL 9017	(x1) Yellow – RAL 1023
83.	IFO Return	Black – RAL 9017	(x1) Green – RAL 6002
84.	HPS	Black – RAL 9017	(x1) Red – RAL 3001
85.	Bitumen	Black – RAL 9017	(x2) Red – RAL 3001

Sl. No:	Service	Base/Ground Colour	Secondary Band Colours
86.	CLO	Black – RAL 9017	(x1) Brown – RAL 8003
87.	VB Tar	Black – RAL 9017	(x2) Brown – RAL 8003
88.	VR AM (Bitumen/VBU Feed)	Black – RAL 9017	(x1) Blue – RAL 5017
89.	VR BH	Black – RAL 9017	(x2) Blue – RAL 5017
90.	VAC. Slop	Black – RAL 9017	(x1) White – RAL 9003
91.	Slop	Black – RAL 9017	(x1) Orange – RAL 2008
92.	Crude Sweet	Dk Grey BS4800 18 B 25	(x1) Red – RAL 3001
93.	Crude Sour	Dk Grey BS4800 18 B 25	(x1) Orange – RAL 2008
94.	VGO/HCU Feed	Oxide Red – RAL 3009	(x1) Signal Grey – RAL 7004
95.	OHCU Bottom/FCCU Feed	Oxide Red – RAL 3009	(x2) Signal Grey – RAL 7004

9.0 PROTECTIVE COATING SYSTEMS

9.1 General

Service / Item	Substrate	Service Temp ^{O}C	Coating System	Comments
Piping, Vessels, Equipment & Un-galvanised structural steel	Carbon Steel	≤ 120	**CS 1**	Tie-coat may be required between primer & build coat
Piping, Vessels & Equipment	Carbon Steel/ Stainless Steel	> 120 < 540	**CS 2**	Primer to be omitted from this system for Stainless Steel substrates
Piping, Heat Exchangers, Vessels & Equipment	Carbon Steel/ Stainless Steel	> 121 < 200	**CS 3**	Primer to be omitted from this system for Stainless Steel substrates
Insulated Piping, Vessels & Equipment	Carbon Steel	> 200	**CS 4**	
Insulated Piping, Vessels & Equipment	Carbon Steel	≤ 200	**INS 1**	Care to be exercised not to apply epoxy phenolic coatings too thickly.
Piping, Vessels & Equipment. Insulated or Un-insulated	Carbon Steel & Stainless Steel	Fluctuating up to 540	**T.S.A.**	Cyclic services to receive corrosion protection using Thermally Sprayed Aluminium T.S.A.
Immersed piling, below jetty deck supports and steelwork in splash zone	Carbon Steel	≤ 65	**CSJ 1**	Above jetty deck, protective coatings to be selected in accordance with other systems in this specification
Un-insulated Piping, Vessels & Equipment	Stainless Steel	> 0 ≤ 120	**SS 1**	Care to be taken when selecting abrasives for cleaning stainless steel.
Internal Lining to Potable water storage tanks	Carbon Steel	≤ 65	**IL 01**	All internal surfaces to be 100% lined. Lining shall be approved for use in contact with potable / fresh water, by a recognised national water industry standard.
Internal Lining to Brackish, Fire or Industrial Water, Crude Oil Storage & Diesel / Fuel Tanks	Carbon Steel	≤ 65	**IL 02**	Crude oil storage tanks with contents that are susceptible to water drop-out shall have base plates (+ 1st 2.4 m up) and underside of roof coated. Brackish, Fire or Industrial Water, Diesel / Fuel tanks to be 100% lined.
Internal Lining to Brackish, Fire or Industrial Water, Crude Oil Storage & Diesel / Fuel Tanks	Carbon Steel	≤ 65	**IL 03**	Crude oil storage tanks shall have base plates (+ 1st 2.4 m up) and underside of roof coated. Brackish, Fire or Industrial Water, Diesel / Fuel tanks to be 100% lined.

Service / Item	Substrate	Service Temp ^{O}C	Coating System	Comments
Galvanised Steel (Duplex system)	Carbon Steel	≤ 120	**GAL 01**	At Contractor request. This coating system to be used when the synergy of two protective coating systems is sought.
Galvanised Steel	Carbon Steel	≤ 120	**GAL 02**	At Contractor request. To be employed when coated for colour purposes.
External surfaces of un-insulated Tanks	Carbon Steel / Stainless Steel	≤ 120	**CS 1**	
Tank bottoms, buried vessels, buried piping	Carbon Steel / Stainless Steel	≤ 90	**UG01**	
Structural steel with concrete fireproofing	Carbon Steel	≤ 120	**INS 1**	Care to be exercised not to apply epoxy phenolic coatings too thickly.
Structural steel/Vessel supports with lightweight fireproofing.	Carbon Steel	≤ 120	**FP01**	

9.2 Make-up of Protective Coating Systems

Coating System	Minimum Surface Preparation	Coating System Make-Up			Min DFT in µm
		Primer Coat	Build Coat	Top Coat	
CS 1	Abrasive Blast Clean to Sa 2½	Inorganic zinc silicate @ 75µm	High build MIO epoxy @ 100µm	Two pack acrylic modified polyurethane @ 50µm	225
CS 2[5]	Abrasive Blast Clean to Sa 2½	Inorganic zinc silicate @ 75µm	Silicone aluminium @ 25µm [2]	Silicone aluminium @ 25µm [2]	125
CS 3[5]	Abrasive Blast Clean to Sa 2½	Inorganic zinc silicate @ 75µm		High temperature silicon acrylic @ 50µm	125
CS 4	Abrasive Blast Clean to Sa 2½	Inorganic zinc silicate @ 75µm			75
INS 1	Abrasive Blast Clean to Sa 2½	Two pack epoxy phenolic @ 150µm		Two pack epoxy phenolic @ 150µm	300
T.S.A.	Abrasive Blast Clean to Sa 2½	Thermally sprayed aluminium shall be applied to carbon and stainless steel surfaces (insulated & non-insulated) when cyclic temperature conditions ≤ 540°C apply.			250[3]
CSJ 1	Abrasive Blast Clean to Sa 2½	Polyamide-cured epoxy primer @ 75µm	Solvent free high solids amine cured epoxy @ 500µm	Solvent free high solids amine cured epoxy @ 500µm	1075
SS 1	Sweep Blast [1]	Silicone aluminium @ 25µm [2]		Silicone aluminium @ 25µm [2]	50
IL01 [4]	Abrasive Blast Clean to Sa 3		2 Pack High Build Pure Epoxy @ 100µm	2 Pack High Build Pure Epoxy @ 100µm	200
IL 02[4]	Abrasive Blast Clean to Sa 3	Modified epoxy phenolic @ 150µm	Modified epoxy phenolic @ 150µm	Modified epoxy phenolic @ 150µm	450
IL 03[4]	Abrasive Blast Clean to Sa 3	Polyamide cured epoxy @ 75µm	High solids amine cured epoxy @ 250µm	High solids amine cured epoxy @ 250µm	575
GAL 01	Sweep Blast [1]	High build epoxy @ 75µm	High build epoxy @ 100µm	Two pack acrylic modified polyurethane @ 50µm	225
GAL 02	Sweep Blast [1]	High build epoxy @ 75µm		Two pack acrylic modified polyurethane @ 50µm	125
UG 01	Abrasive Blast Clean to Sa 2½		Glass Flake Epoxy @ 500µm	Glass Flake Epoxy @ 500µm	1000
FP01	Requirements shall be as per PDRP-8440-SP-0007 “Passive Fire Protection & External Fireproofing of Equipment Support and Structure”				

[1] Sweep blast stainless steel, galvanized steel surfaces using aluminium oxide or garnet abrasive media.

[2] Only air curing heat resistant silicone aluminium paints shall be applied, post heat curing materials are not acceptable.

[3] The DFT for TSA is without sealer coats, where sealer coats are required the DFT must be increased accordingly.

[4] Detailed lining selection shall be the responsibility of the Contractor – see section 1.1.6.

[5] Primer to be omitted for Stainless Steel substrates